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SUPERVISED LEARNING - BINARY CLASSIFICATION

$\{x_1, \ldots, x_n\}$ $\forall i$ $x_i \in \mathbb{R}^d$

$y_1, \ldots y_n$ $\quad y_i \in \{0,1\} / \{-1,1\}$

Goal:

$$h : \mathbb{R}^d \to \{0,1\}$$

Loss / Error

$$\text{Loss}(h) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(h(x_i) \neq y_i)$$

0-1 loss

$$\mathbb{1}(z) = \begin{cases} 1 & \text{if true} \\ 0 & \text{if false} \end{cases}$$

$z =$

$$\min_{h \in \mathcal{H}_{linear}} \sum_{i=1}^{n} \mathbb{1}(h(x_i) \neq y_i)$$

$\mathcal{H}_{linear} = \{ h : h_w(x) = \text{sign}(w^T x) \}$

np-hard problem

$$\text{sign}(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{o/w} \end{cases}$$

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K-Nearest Neighbours

- Given x_{test}, find the k-closest points in the training set - $\{x_1^*, x_2^*, \ldots, x_k^*\}$
- Predict y_{test} = majority $(y_1^*, y_2^*, \ldots, y_k^*)$

DECISION BOUNDARY

Choosing k

- treat as a hyperparameter
- smaller the k, complicated the decision boundary
- solution : cross-validate for k

Issues with K-nn

- Choosing a distance function
- prediction will take computationally long and is expensive
- no model is learnt / cannot throw away data after 'learning'

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DECISION TREES

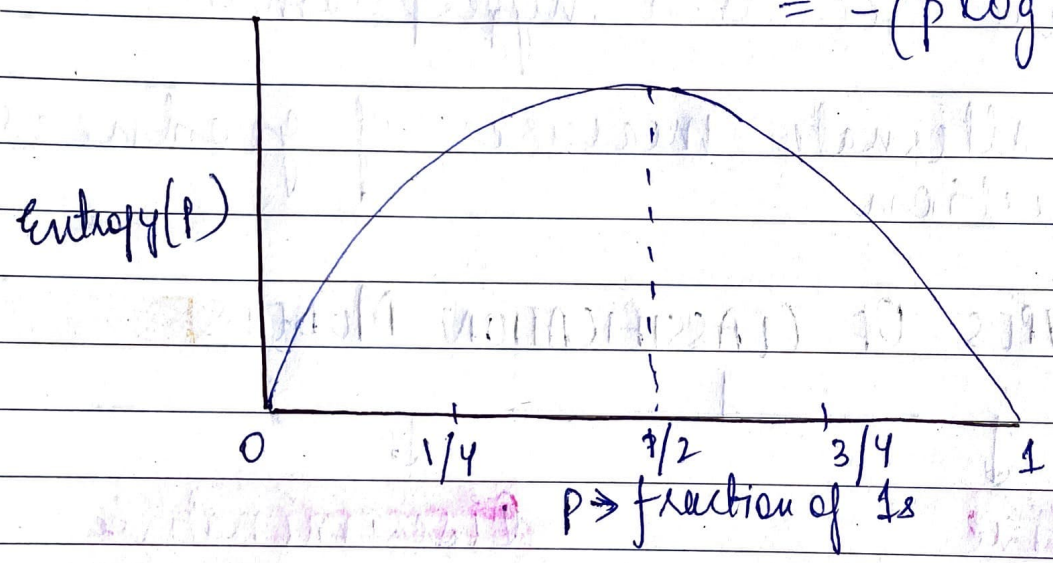
Question: A question is a (feature, value) pair
Eg: height ≤ 180 cm?

How to measure 'goodness' of a question?

- Need a measure of impurity for a set of labels $\{y_1, \ldots, y_k\}$

Entropy

$$\text{Entropy}(\{y_1, \ldots, y_k\}) = \text{Entropy}(p)$$
$$= -(p \log p + (1-p) \log(1-p))$$



Information Gain

Information gain (feature, value) =

$$\text{Entropy}(D) - \left[\gamma\, \text{Entropy}(D_{yes}) + (1-\gamma)\, \text{Entropy}(D_{no})\right]$$

$$\gamma = \frac{|D_{yes}|}{|D|} \rightarrow \text{cardinality}$$

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Algorithm

- Discretize each feature in [min, max] range.
- Pick the question that has the largest information gain.
- Repeat the procedure for D_{yes}, D_{no}

- can stop growing a tree if a node becomes 'sufficiently' pure
- Depth of the tree is a hyperparameter.
- There are alternate measures of goodness of a question

TYPES OF CLASSIFICATION MODELS

Generative

- Models : $P(x, y)$
- modelling feature generation

Discriminative

- Models : $P(y|x)$
- only discriminate label
- Eg : k-nn, decision trees